

Introduction to Online Convex Optimization

Reading Seminar — Sections 5.5–5.8

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1. Randomized Regularization

1.1 Follow-the-Perturbed-Leader

Algorithm 16 — Follow-the-perturbed-leader for convex losses

- 1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
- 2: Let $x_1 = \mathbb{E}_{n \sim \mathcal{D}}[\arg \min_{x \in \mathcal{K}} \{n^\top x\}]$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t .
- 5: Observe f_t , suffer $f_t(x_t)$, and set $\nabla_t = \nabla f_t(x_t)$.
- 6: Update

$$x_{t+1} = \mathbb{E}_{n \sim \mathcal{D}} \left[\arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^t \nabla_s^\top x + n^\top x \right\} \right] \quad (5.4)$$

7: **end for**

- Resembles a **special case of FTRL** (Follow-the-Regularized-Leader).
- **Key difference:** the regularizer is **randomized** — noise $n \sim \mathcal{D}$ is injected into the objective in place of a fixed regularizer.

1.2 Regret Bound for FPL

Theorem (5.8)

Let $\mathcal{D}$ be (σ, L) -stable with respect to the norm $\|\cdot\|_a$. Then FPL (Algorithm 16) attains

$$\text{Regret}_T \leq \eta D(G^*)^2 LT + \frac{1}{\eta} \sigma D,$$

and, optimizing over η ,

$$\text{Regret}_T \leq 2LDG^* \sqrt{\sigma T}.$$

Definition $((\sigma, L)$ -stability)

For $\sigma, L \in \mathbb{R}$, a distribution $\mathcal{D}$ is (σ, L) -stable with respect to $\|\cdot\|_a$ if

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u : \int_n |\mathcal{D}(n) - \mathcal{D}(n - u)| \, dn \leq L \|u\|_a^*.$$

- ...
- ...

Algorithm 17 – FPL for linear losses

1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
2: Sample $n_0 \sim \mathcal{D}$. Let $\hat{x}_1 \in \arg \min_{x \in \mathcal{K}} \{-n_0^\top x\}$.
3: **for** $t = 1$ to T **do**
4: Predict $\hat{x}_t$.
5: Observe the linear loss function, suffer loss $g_t^\top x_t$.
6: Update

$$\hat{x}_t = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} g_s^\top x + n_0^\top x \right\}$$

7: **end for**

- ...
- ...

1.4 FPL for Prediction from Expert Advice

Algorithm 18 – FPL for prediction from expert advice

1: Input: $\eta > 0$.
2: Draw n exponentially distributed variables $n(i) \sim e^{-\eta x}$.
3: Let $x_1 = \arg \min_{e_i \in \Delta_n} \{-e_i^\top n\}$.
4: **for** $t = 1$ to T **do**
5: Predict using expert i_t such that $\hat{x}_t = e_{i_t}$.
6: Observe the loss vector, suffer loss $g_t^\top \hat{x}_t = g_t(i_t)$.
7: Update (w.l.o.g. choose $\hat{x}_{t+1}$ to be a vertex)

$$\hat{x}_{t+1} = \arg \min_{x \in \Delta_n} \left\{ \sum_{s=1}^t g_s^\top x - n^\top x \right\}$$

8: **end for**

- ...
- ...

2. Adaptive Gradient Descent (AdaGrad)

Motivation. RFTL's regret bound (5.7) scales with a regularizer-dependent term, $\max_{u \in \mathcal{K}} \sqrt{2 \sum_t \|\nabla_t\|_t^{*2} \cdot B_R(u \parallel x_1)}$. Which strongly convex R minimizes it? The optimal choice depends on $\mathcal{K}$ and the cost sequence, so we learn the regularizer online: AdaGrad optimizes the regularization (line 5) to shrink the dominant gradient-norm term.

Algorithm 19 — AdaGrad

- 1: Input: parameters $\eta, x_1 \in \mathcal{K}$.
- 2: Initialize: $G_0 = 0$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t , suffer loss $f_t(x_t)$.
- 5: Update $G_t = G_{t-1} + \nabla_t \nabla_t^\top$ and define

$$\text{(diagonal)} \quad H_t = \arg \min_{H \succeq 0, H = \text{diag}(H)} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = \text{diag}\left(G_t^{\frac{1}{2}}\right)$$

$$\text{(full matrix)} \quad H_t = \arg \min_{H \succeq 0} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = G_t^{\frac{1}{2}}$$

- 6: Update $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t$, $x_{t+1} = \arg \min_{x \in \mathcal{K}} \|y_{t+1} - x\|_{H_t}^2$.
 - 7: **end for**
-

3. Bibliographic Remarks

- **Origins.** Regularization for online learning: Grove et al. [2001]; Kivinen & Warmuth [2001].
- **Follow-the-(Perturbed-)Leader.** Kalai & Vempala [2005] coined *follow-the-leader* and analyzed random perturbation as a regularizer (FPL), reviving an idea of Hannan [1957].
- **Follow-the-Regularized-Leader.** *FTRL* coined by Shalev-Shwartz & Singer [2007]; identical *RFTL* by Abernethy et al. [2008]. Hazan & Kale [2008]: RFTL = Online Mirror Descent.
- **AdaGrad.** Duchi et al. [2010, 2011]; diagonal version in parallel by McMahan & Streeter [2010]. Analysis here follows Gupta et al. [2017].
- **Adaptive regularization in deep learning.** AdaGrad, RMSprop [Tieleman & Hinton 2012], Adam [Kingma & Ba 2014]; survey: Goodfellow et al. [2016].
- **Randomization $\leftrightarrow$ regularization.** In special cases, adding randomization $\approx$ deterministic strongly convex regularization [Abernethy et al. 2014, 2016].

4. Exercises

Claim. For any norm $\|\cdot\|$ with dual norm $\|y\|_* := \sup_{\|z\| \leq 1} z^\top y$,

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

Proof. If $x = 0$ (or $y = 0$), both sides equal 0 and the inequality holds. So assume $x \neq 0$ from here on.

Since $x \neq 0$, the unit vector $u = x/\|x\|$ satisfies $\|u\| = 1$, hence it is admissible in the supremum defining the dual norm. Therefore

$$\|y\|_* = \sup_{\|z\| \leq 1} z^\top y \geq u^\top y = \frac{x^\top y}{\|x\|}.$$

Multiplying both sides by $\|x\| > 0$ yields

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

□

Theorem (Name of result)

State the claim here, with math like $R_T = O(\sqrt{T})$.

Proof. Sketch the argument; the QED square is added automatically.

□

Definition (Term)

f is α -strongly convex if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\alpha}{2} \|y - x\|^2.$$

Lemma

Use lemma the same way as theorem.

Idea. Text on the left, matching math on the right.

- Bullet one
- Bullet two

$$x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta \nabla f_t(x_t))$$
$$\Pi_{\mathcal{K}}(y) = \arg \min_{x \in \mathcal{K}} \|x - y\|$$

State the assumption.

State the assumption.

Then

State the assumption.

Then introduce the algorithm.

State the assumption.

Then introduce the algorithm.

Finally, conclude.

A big, centered takeaway statement.

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